

Descriptive Tables in R

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Agenda

- Introduction
 - Descriptive Tables in R
 - Why Are Descriptive Tables Important?
 - What Do Descriptive Tables Include?
- Janitor Package
- dplyr Package
- gtsummary Package
- Class Example
- Conclusion
- References

Descriptive Tables in R

- Summarize and organize data
- Give a simple overview of key data
- Help present results clearly

Why Are Descriptive Tables Important?

- Identify patterns and trends
- Detect inconsistencies or unusual values
- Compare different groups within the data
- Present results in a clear and accessible format

What Do Descriptive Tables Include?

- Categorical variables (e.g., team names)
 - Counts (n)
 - Percentages (%)
- Numerical variables (e.g., number of goals)
 - Mean
 - Median
 - Standard deviation (SD)

Janitor Package

- Used for cleaning and exploring data
- Helps create simple frequency tables
- Function used: `tabyl()`
- Can add percentages for clearer output

Example (Janitor)

```
1 liga %>%  
2   tabyl(Team) %>%  
3   adorn_percentages()
```

When to Use janitor?

- Quick overview of categorical data
- Simple and readable tables
- Early stage of data exploration

Limitations:

- Mainly for simple summaries
- Does not provide advanced statistical measures
- Not intended for final publication tables

dplyr Package

- Core tool for data manipulation and transformation
- Allows flexible and detailed summaries
- Functions like `group_by()` , `summarise()` and `count()`

What Does dplyr Do?

- Group data by categories
- Calculate summary statistics
- Create customized outputs
- Useful for numerical data within groups
- We can calculate multiple statistics at once

Example (dplyr)

```
library(dplyr)

liga %>%
  group_by(Team) %>%
  summarise(
    mean_goals = mean(GF, na.rm = TRUE),
    sd_goals = sd(GF, na.rm = TRUE)
  )
```

When to Use dplyr?

- Need grouped summaries
- More control over calculations
- Preparing data for further analysis

Limitations:

- Output is not automatically formatted
- Tables are not presentation-ready
- Requires more manual coding than janitor

gtsummary Package

- Creates publication-ready descriptive tables
- Automatically summarizes variables
- Produces clean and professional output
- Main function: `tbl_summary()`

What makes gtsummary different?

- Detects variable types automatically
- Applies appropriate summary statistics
- Formats tables for reporting

Example (gtsummary)

```
library(gtsummary)

liga %>%
  select(Team, GF ,GA) %>%
  tbl_summary(by = Team)
```

When to Use gtsummary?

- Preparing results for reports or presentations
- Clean and professional tables required
- Working with multiple variables

Limitations:

- Less flexible than dplyr for custom calculations
- Requires clean and prepared data
- Mainly used at the final stage

Classroom Question

- liga dataset (bundesligR)
 1. How often has each team appeared?
 2. What is the average performance?
 - Points
 - Wins
 - Goals
 3. How to present results clearly?

Solution

Step 1 – Data Preparation

```
library(dplyr)
library(janitor)
library(gtsummary)

liga <- as_tibble(bundesligR)
```

Solution

Step 2 – Frequency Table (janitor)

```
liga %>%  
  tabyl(Team) %>%  
  adorn_pct_formatting(digits = 1)
```

Solution

Step 3 – Summary Statistics (dplyr)

```
liga %>%  
  group_by(Team) %>%  
  summarise(  
    seasons = n(),  
    avg_points = mean(Points, na.rm = TRUE),  
    avg_wins = mean(W, na.rm = TRUE),  
    avg_goals = mean(GF, na.rm = TRUE)  
  ) %>%  
  arrange(desc(avg_points))
```

Solution

Step 4 – Final Table (gtsummary)

```
liga %>%  
  filter(Team %in% c(  
    "FC Bayern Muenchen",  
    "Borussia Dortmund",  
    "Borussia Moenchengladbach"  
  )) %>%  
  select(Team, Points, W, D, L, GF, GA) %>%  
  tbl_summary(  
    by = Team,  
    statistic = list(all_continuous() ~ "{mean} ({sd})"),  
    digits = all_continuous() ~ 1  
  )
```

Conclusion

- Descriptive tables are essential for understanding data
- janitor, dplyr, and gtsummary serve different roles
- janitor: quick overview of categorical data
- dplyr: flexible data analysis and summaries
- gtsummary: clean and publication-ready tables
- Combining these tools improves clarity and reporting

References

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Thank you for your attention